Lab Assignments: Functions in C++

Department: Computer Science and Engineering, IUST Awantipora
Course: Object-Oriented Programming with C++

Lab 1: Basic Function Concepts

1. Write `void printLine(char ch, int n)` and use it to print a border above and below your name.
2. Write `bool isPrime(int n)`. Print all primes between two given numbers.
   Input: `10 30` **Output:** `11 13 17 19 23 29`
3. Write `long long factorial(int n)` and `long long nCr(int n, int r)`, which calls `factorial()`.
   Input: `5 2` **Output:** `10`
4. Write a menu-driven calculator with functions `add`, `subtract`, `multiply` and `divide`. Prototypes go above `main()`, definitions below it. Handle division by zero.
5. **Challenge:** Write `countDigits`, `reverseNumber`, `isPalindrome` and `isArmstrong`.
   Output: `reverseNumber(1234)` → `4321`; `isArmstrong(153)` → `true`

Lab 2: Call by Value

1. Write `void change(int x)` that doubles `x` and prints it. Call it twice from `main()`, then print the original variable.
2. Write `int power(int base, int exp)` that decrements `exp` in a loop. Print `exp` in `main()` before and after the call.
   Input: `2 10` **Output:** `1024`, exp still `10`
3. Write `int gcd(int a, int b)` (Euclid's method) and `int lcm(int a, int b)`. Print the original numbers after the calls.
   Input: `12 18` **Output:** `GCD = 6, LCM = 36`
4. Write `void swapByValue(int a, int b)`. Print the values inside the function and in `main()`. Show that the swap fails.
5. **Challenge:** Write `int collatzSteps(int n)`.
   Output: `collatzSteps(6)` → `8`; `collatzSteps(27)` → `111`

Lab 3: Call by Reference

1. Write `void swapByRef(int &a, int &b)`. Print the values before and after the call.
2. Write `void circleMeasures(double r, double &area, double &circumference)`.
   Input: `7` **Output:** `Area = 153.94, Circumference = 43.98`
3. Write `void sort3(int &a, int &b, int &c)` that sorts three numbers in ascending order.
   Input: `9 2 5` **Output:** `2 5 9`
4. Write `void deposit(double &balance, double amt)` and `bool withdraw(double &balance, double amt)`, driven by a menu.
5. Write `int countVowels(const string &s)`.
   Input: `Kashmir Valley` **Output:** `4`

6. **Challenge:** Write `void normalizeTime(int &h, int &m, int &s)`.
   Input: `1 75 130` **Output:** `2:17:10`

Lab 4: Call by Pointer

1. Print the value and address of a variable in `main()` and inside `void showAddress(int *p)`. Write `void swapByPtr(int *a, int *b)`.
2. Write `bool divide(int a, int b, int *q, int *r)`. Return `false` if `b` is 0 or a pointer is `nullptr`.
   Input: `17 5` **Output:** `Quotient = 3, Remainder = 2`
3. Using pointer arithmetic only, write `int sumArray(const int *p, int n)`, `void reverseArray(int *p, int n)` and `int* findMax(int *p, int n)`.
   Input: `4 8 1 9 3` **Output:** `Sum = 25`, `Reversed: 3 9 1 8 4`, `Max = 9`
4. Write `void arrayStats(const int *p, int n, double *avg, int *maxVal)`. Skip any output passed as `nullptr`.
5. **Challenge:** Write `void reverseStr(char *s)` using two pointers.
   Input: `IUST Awantipora` **Output:** `aropitnawA TSUI`

Lab 5: Student Result Processor (all three methods)

Store the marks of N students in an array in `main()`. Write the functions below and call them from a menu.

Function	Method
`void readMarks(int *marks, int n)`	Pointer
`double average(const int *marks, int n)`	Pointer + value
`char grade(double avg)` (A ≥ 85, B ≥ 70, C ≥ 55, D ≥ 40, F)	Value
`void topper(const int *marks, int n, int &index, int &score)`	Reference
`void applyGrace(int *mark, int grace)` (maximum 100)	Pointer
`void report(const string &title, const int *marks, int n)`	const reference

Input: `78 92 45 66 88`
Output: `Average = 73.8 (B)`, `Topper = student 2 (92)`
After 5 grace marks to students below 50: `Average = 74.8`

Part B: Write `addBonusVal(int m)`, `addBonusRef(int &m)` and `addBonusPtr(int *m)`, each adding 10. Call each on 60 and print the result in `main()`.